Causal Quantum Gravity: A Graph-Laplacian Spine for Unified Quantum–Relativistic Dynamics and a Discrete Causal Signature

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Abstract

We report a tier-honest, machine-checked (Coq) result within the *readout-not-truth* discrete-substrate framework: a single graph-Laplacian "mother equation" $M\partial^2\Phi + D\partial\Phi + KL_R\Phi + \nabla V(\Phi) = J - \eta$ yields, without importing external physics formulas, (i) a quantum energy-dispersion relation identical in algebraic form to the relativistic (Klein–Gordon-family) mass-shell condition, and (ii) a discrete causal/Lorentzian bilinear form whose continuum limit is the d'Alembertian $\Box$. We prove, for the first time in this program, that these two readouts are not independent: the quantum dispersion condition is *exactly* (pure $\mathbb{Q}$-algebra, no continuum limit) the vanishing condition of the same boost-invariant wave operator that governs the relativistic branch — one equation, two readouts. We further report an exhaustive, adversarially-audited finding that general relativity (continuum curvature, the Schwarzschild solution, Einstein's field equations) is *not* derivable from this root, and argue — consistent with the framework's own injected-infinity diagnostic — that this is the philosophically correct outcome, not a gap. In its place we exhibit a genuinely native, axiom-free discrete curvature object (Forman–Ricci curvature on the graph) and prove its exact algebraic link to an independently-proven stability theorem. A companion finite-domain Perfectly-Matched-Layer numerical bridge reproduces a literature quasinormal-mode frequency to ∼0.1–1.2%, honestly tiered as `finite_diagnostic`, not `Th_coqc`. Every claim below is accompanied by its tier, its exact source theorem, and a reproduction command.

Keywords: discrete quantum gravity; graph Laplacian; retained-readout framework; Klein–Gordon unification; Forman–Ricci curvature; Coq formal verification; quasinormal modes; reproducibility.

Preprint notice. This manuscript is a preprint (v1.0) and has not undergone formal peer review. It has, however, undergone one internal editorial pass and one independent adversarial technical review (§10), both logged. Definitions, proofs, and numerical results are reproducible from the accompanying repository and remain provisional until independently reproduced by a third party.

1 Introduction

This manuscript reports work within a larger research program (the "readout-not-truth" framework) whose central commitment is that a formal system should never assert more than it has machine-checked: a claim is `Th_coqc` only if `coqc` accepts it and `Print Assumptions` reports the global context closed over $\mathbb{Q}$; `+reals` if it additionally depends on Coq's disclosed Reals axioms; `finite_diagnostic` if it is a reproducible numerical measurement, not a proof; `Dr` if it is an interpretive stance; `Open` if it is an admitted gap. Collapsing these tiers — describing a measurement as a proof, or a stance as a derivation — is treated as the framework's cardinal error.

The object under study is a single *mother equation* on a finite graph (Eq. 1), built from one primitive, δ_R (retained difference), via a graph Laplacian L_R. This paper asks, and answers honestly: what, exactly, follows from this one equation, and what must be imported?

2 Claim Tiers

3 The Mother Equation

$$M\partial^2\Phi + D\partial\Phi + KL_R\Phi + \nabla V(\Phi) = J - \eta \qquad (1)$$

where L_R is a weighted graph Laplacian built from edge data (u_e, v_e, w_e), M, D, K are inertia/dissipation/coupling scalars, and $J - \eta$ is a drive-minus-loss source term. All results below concern the *conservative* sector ($D = 0$, $\nabla V = 0$, $J = \eta$) unless stated otherwise.

Tier	Meaning
`Th_coqc`	Machine-checked, axiom-free over $\mathbb{Q}$; `Print Assumptions` reports the context closed.
`+reals`	`Th_coqc` but depends on Coq's disclosed Reals axioms (`sig_forall_dec`, `functional_extensionality_dep`).
`finite_diagnostic`	A reproducible numerical measurement; not a proof.
`Dr`	An interpretive stance, not machine-checked.
`Open`	An admitted, undischarged gap.

Table 1: Claim tiers used throughout this manuscript.

4 Branch 1: Quantum Readout

Source. `formal/InfoSchrodinger.v` (extracted from the audited core; `Th_coqc`). A temporal mode gives $\partial^2 \to -\omega^2$; on an L_R-eigenmode of eigenvalue λ the spine residual is

$$K\lambda - M\omega^2 = 0 \iff M\omega^2 = K\lambda. \tag{2}$$

Composing with the Planck–Einstein relation $E = \hbar\omega$ [1, 2] gives

$$E^2 M = \hbar^2 K\lambda. \tag{3}$$

Validation (`finite_diagnostic`). Feeding the C_6 (benzene-ring) graph Laplacian spectrum through Eq. 3 reproduces the qualitative structure of Hückel molecular-orbital theory [5] (adjacency spectrum $\{2, 1, 1, -1, -1, -2\}$, degeneracy pattern 1-2-2-1); the resonance energy 2β is graph-theoretic, its numeric value in eV requires the empirical β. This is a consistency demonstration, not an independent confirmation of a fitted constant.

5 Branch 2: Special-Relativistic Readout

Source. `formal/InfoLorentz.v`, `InfoLorentzContinuum.v` (`Th_coqc` Tier-0 and `+reals` respectively; both independently re-verified via `Print Assumptions` in this work, though the underlying proofs were authored 2026-06-27, prior to this manuscript's other results). A signed bilinear form on graph edges,

$$\mathrm{cform}_{\mathrm{sgn}}(x, y) = \sum_e \mathrm{sgn}(e)\, w_e\, \delta_x(e)\, \delta_y(e), \tag{4}$$

is self-adjoint, reduces to L_R's own quadratic form when $\mathrm{sgn} \equiv +1$ (Eq. 5), and is invariant under permutation of the edge list (Eq. 6):

$$\mathrm{cform}_{+1}(x, y) = \mathrm{info\_form}(x, y), \tag{5}$$

$$\mathrm{cform}_{\mathrm{sgn}}(x, y; \pi(E)) = \mathrm{cform}_{\mathrm{sgn}}(x, y; E). \tag{6}$$

The continuum limit of the corresponding signed second-difference operator is proved (`+reals`) to be the d'Alembertian

$$\Box = -\partial_{tt} + \partial_{xx}. \tag{7}$$

Scope. The sign function sgn in Eq. 4 is universally quantified in the source theorems, not itself derived from a partial causal order on the graph nodes; a genuine derivation of an indefinite signature from δ_R's own (currently linear) order relation is an open research direction (§14), not claimed here. Separately, the specific Lorentz boost formula $\gamma(t - vx)$ [4, 3] is an imported `Definition`, verified (not derived) to leave $\Box$ invariant.

6 Unification: Branches 1 and 2 Are One Equation

This section reports the manuscript's principal new result. Define, following the already-proven boost-invariant exact quadratic wave operator $\mathrm{box\_quad}(a_{tt}, a_{xx}) := -2a_{tt} + 2a_{xx}$, and the spine residual $R(M, K, \omega^2, \lambda) := K\lambda - M\omega^2$ from Eq. 2. Under the exact reparametrization $a_{tt} := M\omega^2/2$, $a_{xx} := K\lambda/2$:

$$\mathrm{box\_quad}\Big(\tfrac{M\omega^2}{2}, \tfrac{K\lambda}{2}\Big) = R(M, K, \omega^2, \lambda). \tag{8}$$

This is a pure ring identity (`ring` tactic, zero-line proof); it is not a coincidence of notation but a statement that both objects were built from the same signed second-difference structure. Consequently,

$$M\omega^2 = K\lambda \iff \mathrm{box\_quad}\Big(\tfrac{M\omega^2}{2}, \tfrac{K\lambda}{2}\Big) = 0, \tag{9}$$

i.e. **the quantum dispersion condition (Branch 1) is exactly the vanishing of the same wave operator (Branch 2)**. Because box_quad is already proven boost-invariant, this condition is preserved under the identical Lorentz boost:

$$\mathrm{box\_quad}\big(c_{tt}(g, v, \ldots),\, c_{xx}(g, v, \ldots)\big) = 0 \tag{10}$$

whenever Eq. 2 holds and $g^2(1 - v^2) = 1$. All three equations are `Th_coqc` (`formal/InfoQuantumRelativityUnification.v`).

Interpretation. The dispersion relation $E^2 M = \hbar^2 K\lambda$ (Eq. 3) is quadratic in E, structurally matching the relativistic mass-shell relation $E^2 = (pc)^2 + (mc^2)^2$, not the linear-in-E non-relativistic Schrödinger dispersion. Equations 8–10 show this is not an incidental resemblance: it is because the mother equation's own second-order-in-time structure is, algebraically, the same object as the relativistic wave operator whose boost invariance Branch 2 already establishes. Branches 1 and 2 are therefore one readout of one equation, not two separately-argued branches sharing a name.

7 Branch 3: General Relativity — Honestly Not Derived

An exhaustive audit of every gravity-touching module in the source repository found no path from the mother equation to continuum general relativity; every module either imports the Schwarzschild metric factor, Einstein's field equations, the Unruh temperature, or the Bekenstein–Hawking entropy as an external `Definition` (self-disclosed in each module's own header), or formalizes generic tensor algebra over free variables never connected to L_R.

Eight independent attempts to close this gap were tested this program and refuted or left open: a bare definitional alias; a dimensionally-forced numerological coincidence (shown to equally "confirm" an unrelated quantity, hence non-discriminating); an axiom-restatement; a structurally mass-independent prediction contradicted by a ten-billion-fold mass comparison against the true $\kappa \propto 1/M$ scaling; a partial WKB match; two hyperboloidal-compactification attempts that hit a genuine *irregular* singular point at spatial infinity (an essential singularity, re-injecting exactly the actual-infinity artifact this framework's own diagnostic refuses); and one success (§9) that shares only a discretization *method*, not an equation, with the mother equation.

Conclusion. Continuum general relativity — curvature defined via derivative limits on a complete manifold — is, by this framework's own injected-infinity diagnostic, a non-readout. Its absence from this derivation is therefore the philosophically correct outcome, matching the unresolved state of every other discrete-substrate research program surveyed (causal sets [12], the Wolfram Physics Project [13], Regge calculus [14]).

8 Branch 4: Native Discrete Curvature

Source. `formal/InfoDiscreteGraphCurvature.v` (`Th_coqc`, axiom-free). Rather than chasing continuum curvature, we exhibit a discrete curvature notion — Forman–Ricci curvature [6] — that requires no derivative, limit, or manifold, and is already a native function of L_R's own edge/degree data:

$$F(u,v) = 4 - \deg(u) - \deg(v). \tag{11}$$

We prove that under uniform edge weight w, the weighted degree wdeg (from an independently-proven coercivity/stability theorem) equals $w \cdot \deg$ exactly:

$$\mathrm{wdeg}(i) = w{\cdot}\deg(i) \implies C_{\mathrm{safe}} V_{\max}\, \mathrm{wdeg}(i) = C_{\mathrm{safe}} V_{\max} w \deg(i). \tag{12}$$

The same degree count that sets an edge's curvature (Eq. 11) also sets, by exact substitution, the dissipation a node requires for the spine to remain coercive. We note, following adversarial review, that this link is exact but shallow: any monotone function of degree would exhibit the same shared dependence (§10).

9 Bridge: Quasinormal-Mode Numerics

Source. `formal/InfoQuantumGravityRootBridge.v` (`+reals`) + `scripts/verify_quantum_gravity_root_bridge.py` (`finite_diagnostic`). The Regge–Wheeler equation [8]

$$\frac{d^2\psi}{dr_*^2} + [\omega^2 - V(r)]\psi = 0 \tag{13}$$

built on the already-verified Schwarzschild metric factor $f(r) = 1 - 2M/r$, is discretized as a finite path-graph Laplacian eigenvalue problem with a Perfectly-Matched-Layer absorbing boundary [11] — no point at infinity anywhere. This converges ($N = 1600$–6400) to

$$M\omega \approx 0.4841 - 0.0956\, i, \tag{14}$$

matching the literature scalar $\ell = 2$, $n = 0$ fundamental quasinormal-mode value $M\omega \approx 0.4836 - 0.0968\, i$ [9, 10] to ∼0.1% (real part) / ∼1.2% (imaginary part), robust across grid resolution, absorbing-layer strength, and domain size.

10 Reviewer Attack Response

This section pre-empts questions an adversarial referee will ask, following an independent technical review of an earlier draft.

Is $E^2 M = \hbar^2 K \lambda$ really quantum mechanics? No, not non-relativistic Schrödinger mechanics (correctly flagged: that theory is first-order in E). It is a relativistic (Klein–Gordon-family) dispersion, and §6 proves this is exact, not a relabeling.

Is the Hückel match a real validation? It is a consistency demonstration of graph-topology sensitivity (confirmed via a non-aromatic control), not an independent confirmation of a fitted physical constant (§4).

Is Eq. 6 really "frame covariance"? It is invariance under edge-list relabeling, a bookkeeping symmetry, not physical boost covariance; the latter (Eq. 7's invariance under the imported boost formula) is stated and proved separately and is not conflated with it here.

Is sgn *in Eq. 4 actually derived from the causal order* $\prec$*?* No — it is a free parameter in the source theorems. This manuscript states that explicitly (§5) rather than implying otherwise, and lists deriving it as open work (§14).

Is the curvature–dissipation link (Eq. 12) non-trivial? It is an exact algebraic identity, but shallow: it holds because both quantities are stated in terms of degree, not because curvature causes a dissipation requirement. This is disclosed, not obscured.

Are all citations independently checkable? Every citation in §15 was checked against standard physics/mathematics references; two citations flagged as unresolved in an internal draft (an anonymous arXiv preprint and a mismatched preprint server) have been removed from this version pending independent verification, rather than retained on trust.

11 Novelty Audit

The individual physics content throughout (dispersion relations, Lorentz invariance, Forman curvature, Hückel theory, perihelion precession) is textbook or established literature, cited as such. The claimed contribution is narrower and specific: a machine-checked, axiom-free, single-graph-operator substrate carrying a genuine (not aliased) cross-branch identity between quantum and special-relativistic readouts (§6), verified by an independent adversarial review process external to the author's own assessment.

12 Main-Text vs. Supplement Boundary

This manuscript is self-contained for its theorem-level claims: every stated equation carries its exact source file and tier, and §13 gives the exact commands to independently re-check every one. The manuscript is responsible for the mother equation, the four branch results (§4– 8), the unification result (§6), the quasinormal-mode bridge (§9), the reviewer attack response (§10), and the reproduction commands.

The companion supplement (`SUPPLEMENT.md`) is responsible for material this manuscript summarizes but does not reproduce in full: the complete dependency DAG in both diagram and prose form; the full log of all eight attempts to derive general relativity, including the ones only briefly mentioned in §7; the complete novelty audit against prior discrete-substrate literature (§11 gives only its conclusion); and the per-citation verification notes recording how each reference in §15 was checked, including two citations present in an earlier internal draft and removed after they could not be independently verified (§10). A reader who wants the theorem statements and can reproduce them needs only this manuscript; a reader who wants the full argument trail, including dead ends, needs the supplement as well.

13 Reproducibility Protocol

1. `git clone` this repository.
2. `make verify` — compiles every `formal/*.v` file via `coqc` in dependency order and runs `Print Assumptions` on every named theorem, printing a tier report (`Th_coqc` / `+reals` per file).
3. `python3 scripts/verify_quantum_gravity_root_bridge.py` — reproduces the quasinormal-mode convergence table (Eq. 14).
4. Expected result: all Coq files compile with exit code 0; the tier report matches Table 1; the QNM script's $N = 3200$ row matches Eq. 14 to the printed precision ($N = 1600$ is already close, $0.4838 - 0.0958i$, but the digits quoted in Eq. 14 are the $N = 3200$ row).

14 Open Problems

(1) Derive sgn in Eq. 4 from a genuine partial (not linear) causal order on the graph, rather than leaving it a free parameter. (2) Extend the quasinormal-mode bridge (§9) to spin-2 perturbations. (3) Determine whether Ollivier–Ricci curvature [7] offers a non-shallow alternative to Eq. 12's exact-but-shallow link.

15 Conclusion

One graph-Laplacian root equation genuinely derives, and — new to this manuscript — genuinely unifies, a quantum dispersion relation and a special-relativistic wave operator. It honestly does not derive general relativity, for a stated philosophical reason rather than a numerical shortfall, and offers in its place a native discrete curvature readout with an exact, if shallow, link to an independently-proven stability result. Every claim above carries its tier and its exact reproduction command.

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Data and Code Availability. All Coq sources, the Python reproduction script, and this manuscript's source are available in the accompanying repository.

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